instrumentation, Valgrind is definitely the platform for you. It allows extensions to the core Valgrind platform itself, as well as building tools on top of it. It has a very active user and developer community.

Approaches in adding instrumentation code

There are two major approaches in DBI when adding the instrumentation code. One approach is what is known as a 'Dis-assembly and Re-synthesise' (D & R) approach, where the binary executable is first disassembled, analysed, an intermediate representation is built, on top of which the instrumentation code is added, and then the complete code containing both the original application and the instrumentation code is lowered to the machine level. The other approach is what is known as 'Copy and Annotate' (C & A), wherein the incoming instructions of the application binary are copied through, as is. Each incoming instruction is annotated with its effects via data structures, as in DynamoRIO, or through an instruction querying API, as in Pin. Instrumentation tools use the annotations to guide the instrumentation. Instrumentation code is inter-leaved with the original application binary without causing any perturbation.

Valgrind uses the D & R approach, whereas Pin and DynamoRIO use the C & A approach. Each approach has its own advantages and disadvantages. D & R is heavy-weight, and can add to runtime execution overhead. C & A is lightweight, and can be faster. However D & R allows arbitrary instrumentation, which may not be possible with C & A. D & R also allows combined optimisation of both application and instrumentation code, and is much closer to a dynamic optimisation system. A detailed discussion of Valgrind's DBI framework can be found in *http://www.valgrind.org/docs/valgrind2007.pdf.*

My 'must-read book' for this month

This month's 'must-read book' suggestion comes from one of our readers, M Kumaraswamy, who recommends the book 'Hadoop: The Definitive Guide' by Tom White. Hadoop is an open source Apache project (*www.hadoop.org*). Kumaraswamy claims that every programmer must know about Hadoop, whether you use it or not. Most of the Internet's large distributed programming environments have been based on Hadoop. If you are a student looking to write a data-intensive distributed application, Hadoop is obviously the framework to choose. It is based on Google's MapReduce framework and the Google File System. It is an interesting architecture to study, understand and experiment with.

Tom White's book contains detailed descriptions of the Hadoop Distributed File System (HDFS), MapReduce, Hadoop Cluster and Hadoop's database, HBase. In short, it teaches you how to build a complex distributed system for data processing. Thank you, Kumaraswamy, for your suggestion that we should discuss Hadoop in detail in this column. We will definitely do that.

If you have a favourite programming book that you think is a must-read for every programmer, please do send me a note with the book's name, and a short write-up on why you think it is useful, so I can mention it in this column. That would help many readers who want to improve their coding skills.

If you have any favourite programming puzzles that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at *sandyasm_AT_yahoo_DOT_com*. Till we meet again next month, happy programming, and here's wishing you the very best! END

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